

Dept. of CCE
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University

Mid: I

Semester: 7th

Batch : 14th Marks 15

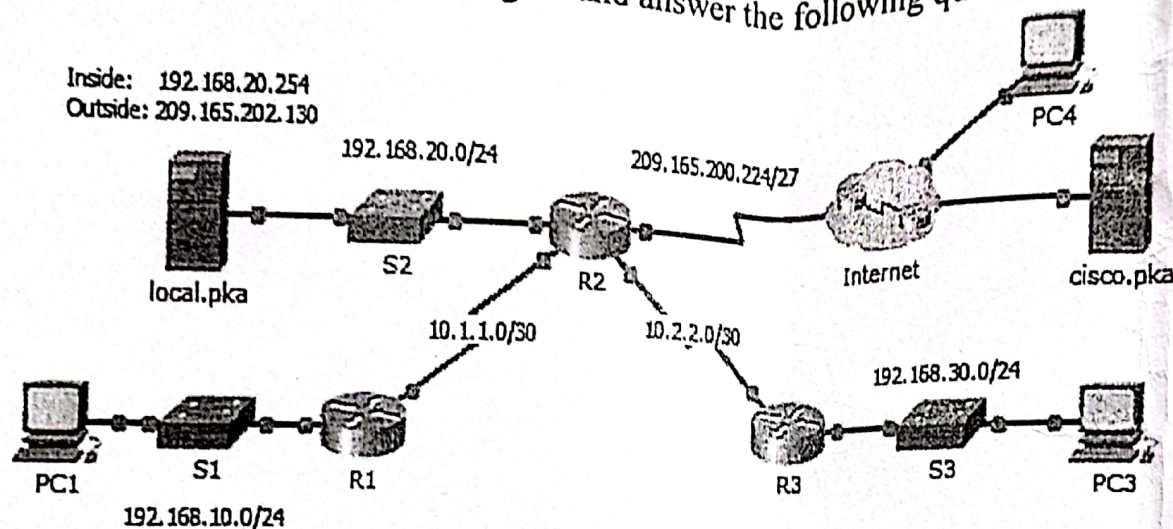
Course Code: CCE 415

Time: 30 Min Course Title: Network Routing and Switching Total

1. What two tasks do dynamic routing protocols perform?
2. What is a disadvantage of using dynamic routing protocols?
3. Which dynamic routing protocol was developed as an exterior gateway protocol to interconnect different Internet providers?
4. Write two statements regarding classless routing protocols?
5. What is the purpose of the passive-interface command?
6. Having configured RIPv2 on an enterprise network, an engineer enters the command network 192.168.10.0 into router configuration mode. What is the result of entering this command?
7. A destination route in the routing table is indicated with a code D.
Which kind of route entry is this?
8. Which two requirements are used to determine if a route can be considered an ultimate route in a router's routing table?
9. Which route is the best match for a packet entering a router with a destination address of 10.16.0.2?
10. Which type of route requires a router to perform a recursive lookup?

10X1.5=15

1. Draw PSTU campus network topology diagram from core to access point mentioning required router, switch, server, fiber, Ethernet, Link aggregation. Consider the basic needs of Campus Network. 5
2. Analyze the given topology in the figure and answer the following question. 10



A: Configure Dynamic NAT with PAT

- i. On R2, configure a standard ACL named R2NAT that uses three statements to permit, in order, the following private address spaces: 192.168.10.0/24, 192.168.20.0/24, and 192.168.30.0/24.
- ii. Configure R2 with a NAT pool named R2POOL that uses the first address in the 209.165.202.128/30 address space. The second address is used for static NAT later.
- iii. Associate the named ACL with the NAT pool and enable PAT.
- iv. Configure R2 interfaces with the appropriate inside and outside NAT commands.

B: Configure Static NAT

- i. Create a static NAT translation to map the local.pka inside address to its outside address.
- ii. How to view the NAT translations on R2.

Dept. of Computer and Communication Engineering
 Patuakhali Science and Technology University

Mid Examination of B.Sc. Engg. (CSE), Level: IV Semester: I Session: 2017-2018
 Course Code: CCE-415 Course Title: Network Routing and Switching Sessional
 Credit Hour: 3.00 Full Marks: 15 Duration: 55 mins

1. Create a static route topology with three routers so that the routers see all networks and also answer the following question. Assume the following situation: 5.5

The Lab_A router is connected to two networks, 172.16.10.0 and 172.16.20.0. On the Lab_B router, you have direct connections to networks 172.16.20.0 and 172.16.30.0. You need to add routes to networks 172.16.10.0 and 172.16.40.0. On router Lab_C, create a static route to networks 172.16.10.0 and 172.16.20.0, which are not directly connected. Create static routes so that router Lab_C can see all networks.

- You need to add routes to networks 172.16.30.0 and 172.16.40.0 to Lab_A router. Write configuration command line.
- You need to add routes to networks 172.16.10.0 and 172.16.40.0 to Lab_B router. Write configuration command line.
- How to create static routes so that router Lab_C can see all networks. Write configuration command line.
- What is privileged mode and enabled mode?

2. What command was used to generate the following output? Explain the following output. 2

Codes: L - local, C - connected, S - static, [output cut]

10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks

C 10.0.0.0/8 is directly connected, FastEthernet0/3

L 10.0.0.1/32 is directly connected, FastEthernet0/3

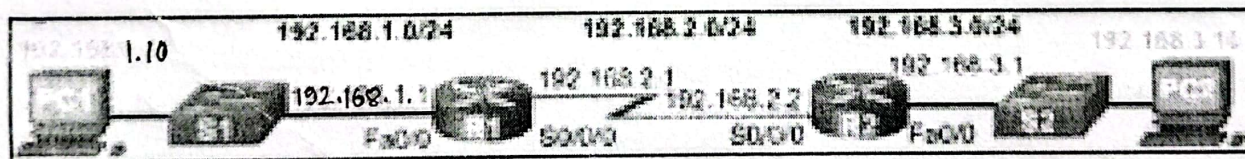
C 10.10.0.0/16 is directly connected, FastEthernet0/2

L 10.10.0.1/32 is directly connected, FastEthernet0/2

C 10.10.10.0/24 is directly connected, FastEthernet0/1

L 10.10.10.1/32 is directly connected, FastEthernet0/1

3.



What are the components of document network addressing? Write the document network addressing of the above topology. How do you configure basic settings on the router R1 and R2 to route between two directly connected networks, using CLI? 4

4. Write the main function of router. How to determine the best path, when the router searches its routing table for a network address that matches the destination IP address of the packet? 3.5

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Final Examination of B. Sc. Engineering in CSE Level: 4 Semester: I Session: 2017-2018

Course Code
CCE-415

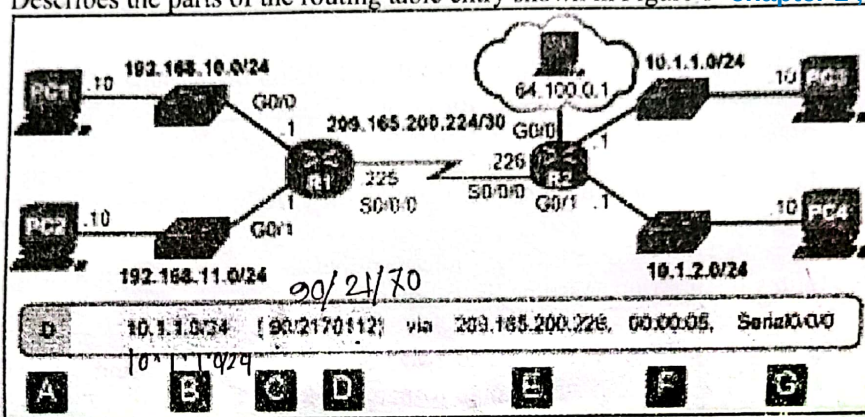
Course Title
Network Routing and Switching

January-June 2021

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- ✓ 1 [A.] "Routers Are Computers" justify the statement with explanation. Write some Router vendors. 4
name and also which is popular to networking community. [chapter 1 - slide - 10](#)
[B.] How do you connect devices for a small, routed network? [chapter - 1, slide - 2](#) 4
[C.] Describes the parts of the routing table entry shown in Figure 1 [chapter 1, slide - 56](#) 3

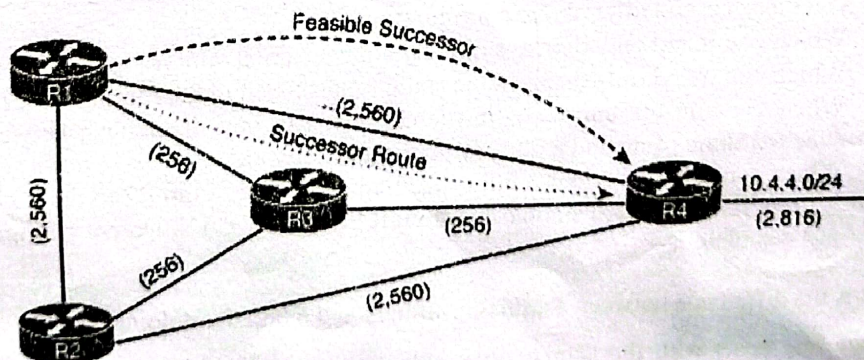


- [D.] What happens if a routing table has two or more paths with identical metrics to the same destination network? [chapter - 1, slide - 51](#)

- 2 [A.] Write the purposes of dynamic routing protocols in a computer network. [chap - 3, slide - 2](#) 3
[B.] Can you explain what IPv6 is and why it's important? Why do we need NAT when using IPv6? 2
[C.] A router must find the best match in the routing table. But how does it do that? [book page 268](#) 3
[D.] Write the static and dynamic routing advantages and disadvantages. [book page 236, 237](#) 3
[E.] Differentiate between Enhanced Interior Gateway Routing Protocol (EIGRP) and Border Gateway Protocol (BGP). 3

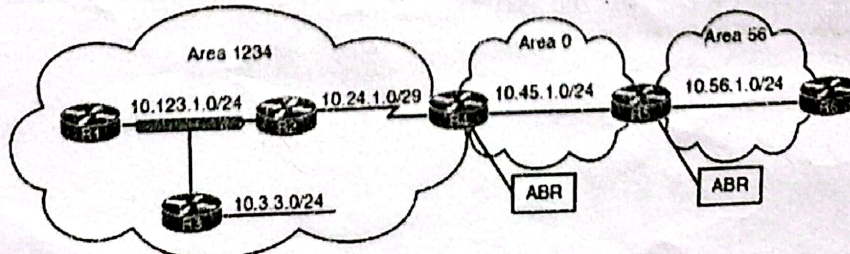
- 3 [A.] Defines important terms of below topology related to EIGRP and correlates them to Figure 2. 4

[Chapter EIGRP.pdf - page - 5](#)



- [B.] How do you configure OSPF of below Topology Figure 3

[Chapter OSPF.pdf, page - 13](#)



[C.] What do you mean by BGP Message Types? Explain the different Message Types in BGP in detail.

[D.] Explain the Path selection criteria used in BGP.

Base Protocol de Sa Unit

4 [A.] What is STP and Redundant Links? How STP works?

[B.] Suppose there are links between switches S₁ and S₂, and S₁ and S₃. Switch S₂ and S₃ are connected with 4 PCs with different VLANs (PCs with same VLAN maintain same group name). The links are configured to transmit traffic coming from VLANs 10, 15, 20, 30, and 99 across the network. Design the topology. What should do to active the function of that network? [book page - 400](#)

[C.] What is BPDU? What happens to a switch's ports when it receives a BPDU from another switch that has a lower priority or an equal priority?

[D.] Draw a flowchart to troubleshoot missing VLAN. [book page 428](#)

5 [A.] The store-and-forward method makes a forwarding decision on a frame after it has received the entire frame and checked the frame for errors. Which mechanism is used? Justify with proper figure and explanation. [book page 316](#)

[B.] How to handle incoming data frames by maintaining the MAC address table? Briefly describe with figure. [book page - 315](#)

[C.] Create a static route topology with three routers so that the routers see all networks and also answer the following question. Assume the following situation:

2.2.2.4 Packet Tracer - Configuring IPv4 Static and Default Routes Instructions.pdf

The Lab_R1 router is connected to two networks, 172.31.1.0 and 172.31.1.192. On router Lab_R2, create a static route to networks 172.31.1.0 and 172.31.1.192, which are not directly connected. Router Lab_R2 has direct connection with 172.31.1.196 and one other network mentioned in the situation. Create static routes so that router Lab_R2 can see all networks. On the Lab_R3 router, there have direct connections to 172.31.0.0 and to two common networks. You need to add routes to networks 172.31.1.0 and 172.31.1.128 of respective routers.

- How many static routes are required by each router to reach networks that are not directly connected?
- Test connectivity to the R2 and R3 LANs by pinging PC2 and PC3 from PC1. Why were you unsuccessful?
- Why does a recursive static route require two routing table lookups?
- Configure a recursive static route to every network not directly connected to R1, including the WAN link between R2 and R3.
- Test connectivity to the R2 LAN and ping the IP addresses of PC2 and PC3. Why were you unsuccessful after above configuration?
- Which command only displays directly connected networks?
- Which command only displays the static routes listed in the routing table?
- When viewing the entire routing table, how can you distinguish between a directly attached static route and a directly connected network?
- Which command provides a fully specified static route from R2 to the R3 LAN?
- Write a fully specified static route from R2 to the R1 LAN. Do not configure the route; just calculate it.

✓ 6 [A.] What is the difference between a collision domain and a broadcast domain? [book page - 292](#)

[B.] Assume the hosts with the source IPv4 addresses (192.168.10.10 (PC1) and 192.168.11.10 (PC2)) and 192.168.12.10 (PC3)) send packets requesting a connection to the server at the public IPv4 address (209.165.200.254). Draw a topology that illustrates the NAT translation process between three clients and the web server. Explain the traffic flow from inside to outside and outside to inside based on dynamic NAT process. [book page - 661 , 662](#)

[C.] Is static routing a protocol? How do you configure IPv4 Fully Specified Static Route to provide a backup connection? Explain with topology. [book page - 147 , 167 ,](#)

Final Examination of B. Sc. Engineering in CSE Level: IV Semester: I Session: 2016-2017

Course Code
CCE-415

Course Title
Network Routing and Switching

January-June 2020

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- | No. | Serial | Questions | Marks |
|-----|--------|---|-------|
| 1 | [A.] | State the learning outcomes of the network routing and switching course in terms of getting a job as a network engineer. | 3 |
| | [B.] | <p>i) Note down the primary function of a router. chap - 1, slide - 13</p> <p>ii) Write two statements that correctly describe the concepts of administrative distance and metric of a router. chap - 1, slide 56 / exercise - 3</p> <p>iii) For packets to be sent to a remote destination, mention three pieces of information must be configured on a host. chap - 1, slide - 21, exercise - 4</p> <p>iv) Consider the following routing table entry for R1: D 10.1.1.0/24 [90/2170112] via 10.2.1.1, 00:00:05, Serial0/0/0. What is the significance of the Serial0/0/0? chap - 1, slide - 56, exercise - 12</p> <p>v) A network administrator configures interface G0/0 on R1, with the ip address 172.16.1.254 255.255.255.0 command. However, when the administrator runs the show ip route command, the routing table does not show the directly connected network. What is the possible cause of the problem? chap - 1, exercise - 14</p> | 5 |
| | [C.] | Differentiate between switch, hub, repeater and router. | 3 |
| | [D.] | How does the router work? Explain with a basic network. book page - 51 | 3 |



Examine the network requirements.

You have been given the 172.16.254.0/24 address space to use in your network design. The network consists of the following segments:

- > The network connected to router R1 will require enough IP addresses to support 20 hosts.
- > The network connected to router R2 will require enough IP addresses to support 20 hosts.
- > The link between router R1 and router R2 will require IP addresses at each end of the link.

Assign subnetwork addresses to the topology diagram

- a. Assign subnet 1 to the network attached to R1.
- b. Assign subnet 2 to the link between R1 and R2.
- c. Assign subnet 3 to the network attached to R2.

Assign appropriate addresses to the device interfaces.

- a. Assign the first valid host address in subnet 1 to the LAN interface on R1.
- b. Assign the last valid host address in subnet 1 to PC1.
- c. Assign the first valid host address in subnet 2 to the WAN interface on R1.
- d. Assign the last valid host address in subnet 2 to the WAN interface on R2.
- e. Assign the first valid host address in subnet 3 to the LAN interface of R2.
- f. Assign the last valid host address in subnet 3 to PC2.

Document the addresses to be used in Table 1 and design the network with IP

Device	Interface	IP Address	Subnet Mask	Default Gateway

- i. How many subnets are needed for this network?
- ii. What is the subnet mask for this network in dotted decimal format?
- iii. What is the subnet mask for the network in slash format?
- iv. How many usable hosts are there per subnet?
- v. From the host attached to R1, is it possible to ping the default gateway?
- vi. From the host attached to R2, is it possible to ping the default gateway?
- vii. From router R1, is it possible to ping the Serial 0/0/0 interface of R2?
- viii. From router R2, is it possible to ping the Serial 0/0/0 interface of R1?
- ix. What is the status of the FastEthernet 0/0 interface of R1?
- x. What is the status of the Serial 0/0/0 interface of R1?

- xi. What is the status of the FastEthernet 0/0 interface of R2?
- xii. What is the status of the Serial 0/0/0 interface of R2?
- xiii. What are the routes present in the routing table of R1?
- xiv. Are the routes present in the routing table of R2?
- xv. Are there any devices on the network that cannot ping each other?
- xvi. What is missing from the network that is preventing communication between these devices?

- 3
- [A.] Summarize the various ports and cables that are required for configuring the router of a computer network. [chap - 1, slide 24, 25](#) (3)
- [B.] Write down the classifications of routing protocol with a flowchart. What are the advantage and disadvantage of distance vector routing protocol. [eigrp slide - 1](#) (4)
- [C.] i) What are the two tasks of dynamic routing protocols perform? [chap - 3, exercise - 1](#) (4)
 ii) What is the purpose of the passive-interface command? [chap 3, exercise - 5](#)
 iii) Having configured RIPv2 on an enterprise network, an engineer enters the command network 192.168.10.0 into router configuration mode. What is the result of entering this command? [chap 3, exercise - 6](#) (2)
 iv) Write down two requirements that are used to determine if a route can be considered an ultimate route in a router's routing table? [chap 3, exercise - 8](#) 3 3
- [D.] How does RIP routing protocol work in a network router? [book page - 231](#) 3
- 4 [A.] What is the purpose and function of NAT? Design NAT Topology Showing NAT Address Types examples. [book page - 632 + 639](#) 5 3
- [B.] Traffic between two switches is slow. You issue the show interface command on the uplink between the two switches and you see the following: [chapter 5 - ?](#) 3
- output omitted!
 0 input packets with dribble condition detected
 180749 packets output, 8004302 bytes. 0 underruns
 0 output errors, 45345 collisions, 0 interface resets
 0 babbles, 45345 late collisions, 0 deferred
 0 lost carrier, 0 no carrier
 0 output buffer failures, 0 output buffers swapped out
- [C.] What are several possibilities for this problem?
 Is static routing a protocol? How do you configure IPv4 and IPv6 floating static route to provide a backup connection? [book page - 188, 192](#) 6
- 5 [A.] What are the advantages and disadvantages of static routing? [book - 236](#) (3)
- [B.] Host A wants to send data to host B. Host B is on a different segment from host A. The two segments are connected to each other through a router. What will host B see as the source MAC address for all frames sent from host A? [chap - 1 - ?](#) 3
- [C.] Suppose there are links between switches S₁ and S₂, and S₁ and S₃. Switch S₂ and S₃ are connected with 4 PCs with different VLANs (PCs with same VLAN maintain same group name). The links are configured to transmit traffic coming from VLANs 10, 15, 20, 30, and 99 across the network. Design the topology. [book page - 400](#) 4
- [D.] What should do to active the function of that network?
 How do you Configuring SSH? Write down configuration steps and command example with topology. [book page - 358](#) (4)
- 6 [A.] What is tagging for VLAN? [book page - 403](#) 2
- [B.] What will an Ethernet switch do if it receives a unicast frame with a destination MAC that is listed in the switch table? [book page 314](#) 3
- [C.] How do you configure PAT with a single address topology example? [book page - 669](#) 3
- [D.] Briefly analyze PAT process (PCs to servers and Servers to PCs) with previous PAT example 6(c). [book page - 670](#) 6

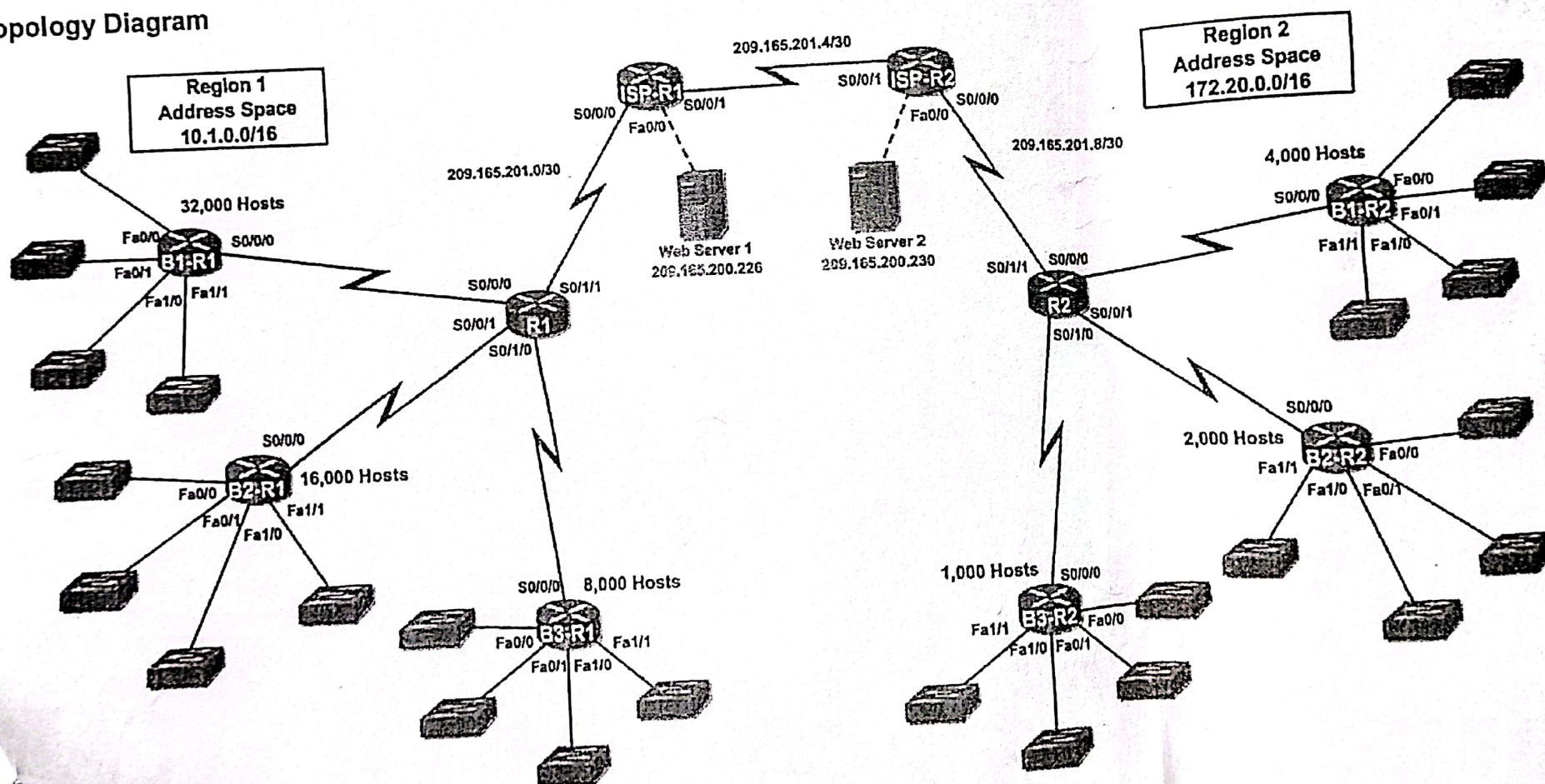
1. Viva Voce 15
2. Using the topology and the following requirements, design an addressing scheme: - (Topology attached in opposite page) 30
The WAN links between R1 and R2 and their respective ISP routers are already configured. Also, the links between the ISPs and the Web Servers are already configured.
The address space for Region 1 is 10.1.0.0/16. Each branch router (B1-R1, B2-R1, and B3-R1) should be allotted address space based on the following requirements. Starting with the largest requirement, assign address space to each router.
B1-R1 needs space for 32,000 hosts.
B2-R1 needs space for 16,000 hosts.
B3-R1 needs space for 8,000 hosts.
Divide the address space for each branch router into four equal subnets. Record the subnets in the table below.
For the WANs in Region 1, subnet the address space 10.1.255.240/28. Record the subnets in the table below.
The address space for Region 2 is 172.20.0.0/16. Each branch router (B1-R2, B2-R2, and B3-R2) should be allotted address space based on the following requirements. Starting with the largest requirement, assign address space to each router.
B1-R2 needs space for 4,000 hosts.
B2-R2 needs space for 2,000 hosts.
B3-R2 needs space for 1,000 hosts.
Divide the address space for each branch router into four equal subnets. Record the subnets in the table below.
For the WANs in Region 2, subnet the address space 172.20.255.240/28. Record the subnets in the table below.
3. Lab Problem Solved 10
4. Coursera Certificate 05
5. Project Work 10

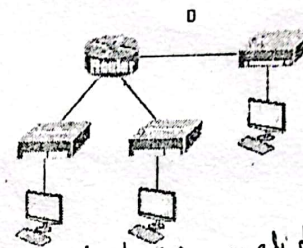
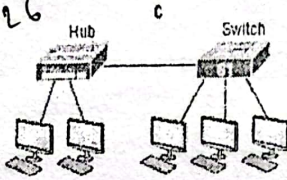
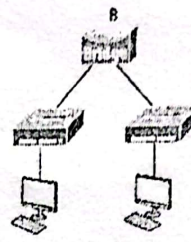
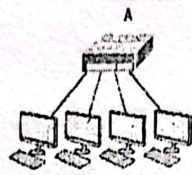
1.

30 Marks

$n \times 10^2 N/2$

Topology Diagram





255.255.255.0
 192.100.0.0 /26
 sub net mask
 last address
 is used by
 192.100.0.63

मार्क 3 व
 VLSD 20
 मर्क

Ans → network
 last is boundary

total host 64 valid host = 64 - 2 =

- 3 B a. What is the 127.0.0.1 address used for? How do you find the network address from a listed IP address? 7
- b. In your own words, explain the difference between DHCP's static and dynamic address allocation options.
- 4 A a. Define subnetting. How to create subnets? Describe the benefits of subnetting. 7
- b. What is default mask for a address of class A in CIDR notation? Write down the subnet mask for CIDR value /9, /12, /16, /30. 255.0.0.0
- 4 B a. Implement a class C /26 (with three networks) How many hosts are available there? 7
- b. Complete the following based on the decimal IP address.

Decimal IP Address	Address Class	Number of Subnet and Host Bits	Number of Subnets	Number of Hosts
10.25.66.154/23		9		512
172.31.254.12/24		8		256
192.168.20.123/28		4		16
63.24.89.21/18		14		16,384
128.1.1.254/20		12		4,096
208.100.54.209/30		2		4

2-12
 10.25.66.154
 172.31.254.12
 192.168.20.123
 63.24.89.21

24
 24,000

- 5 A What is Variable Length Subnet Masks? What are the differences between typical classful and classless network design? Discuss with a suitable example. 7
- 5 B Design a network using VLSM with 11 subnetworks, two block sizes of 64, one of 32, five of 16, and three of 4. 7
- 6 A Which algorithm is used by OSPF to construct a shortest path tree? Describe the OSPF Operations (Neighbor and adjacency initialization, LSA flooding, SPF tree calculation). 7
- 6 B a. When do we use NAT? Identify the basic operations of NAT. 7
- b. Write down the advantages and disadvantages of implementing NAT. Describe about various types of Network Address Translation.

allows
 24
 24,000